

Application on joint species distribution model

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Scaling up from GLMs to GLMMs

Joint species distribution models are just glorified generalised linear mixed-effect models (GLMMs)

Single-species GLM

The response of species A to environment X across sites.

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Whiteboard: graphical way

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$$Y \sim \text{Binomial}(N, p)$$

$$\text{logit}(p) = \alpha + \beta X$$

Single-species GLM

The response of species A to environment X across sites.

$$Y \sim \text{Bernoulli}(p)$$

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Single-species GLM

The response of species A to environment X across site i .

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Whiteboard: expand to multi-species dataset

Multi-species GLMM

The response of species j to environment X across site i .

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Multi-species GLMM

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y ~ (1 | site) + (1 + X | species) + X,  
family = binomial("logit")
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Detour 1: incorporating traits

How do traits moderate the response of species j to environment X across site i .

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Whiteboard: [link to multivariate ordination](#)

Fitting JSMD (finaaaaally)



- Recommended packages (with traits and latent variables)
 - `boral` (Bayesian)
 - `gllvm` (Frequentist)

Fitting JSDM (finaaaaally)



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 - Transferable skill beyond JSDMs
 - Fast
 - Easy-to-learn syntax

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Pair up with your laptops

<https://github.com/hrlai/bayes-workshop>

Coding session



Run `code/part2.R` up until the `brm` section.

Model diagnostics



*Sometimes classical statistics gives up.
Bayes never gives up ... so we're under more responsibility to check our models.*

— Andrew Gelman

Post-processing posteriors



You get posterior for everything.

This is one of the perks of Bayesian inference.

Whiteboard: why we should calculate everything across posteriors

Bonus: information borrowing

Parameters can be correlated;
this can be a good thing called *shrinkage*.

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- When you spend too much time building ever more complex Bayesian models, it's probably a good time simplify your questions instead
- At which point you may find that you don't need Bayes anyways
- I still use frequentist approach, and everytime I return to it, I realise how much more I appreciate frequentist because of what I have learnt from Bayesian inference.

Thank you!

Feedback welcome! Link to Google Form below:



On the GitHub repository:

- Further readings
- Suggested R packages